

Midterm Exam

CMU 10-701: Introduction to Machine Learning (Fall 2022)

Oct 12, 2022, 10:10 - 11:30 AM (80 minutes)

Name: Laura McDonnell
Andrew ID: lmcdonne

Instructions: Please read this before your start!

- **IMPORTANT!** Before you start, write your name and Andrew ID in the box above.
- This exam has 25 pages, including this cover page and the last page (scratch paper). Please look over the exam first. **If there is a missing page, raise your hand now.** This exam contains 7 parts. The total number of points is 80.
- **Write your answers in the designated boxes for each question.** The boxes should be much larger than enough.
- Please write all answers in pen.
- You are allowed to bring one double-sided sheet of hand-written cheat sheet. No electronic devices, books or lecture notes may be used during the exam.
- Don't stop! If you are stuck, move on to the next question and you can come back later. Every part has some easy partial credits you can grab.
- You have 80 minutes to complete the exam. Good luck!

Overview

- Part 1: Multiple Choice Questions (20 pts)
- Part 2: Decision Trees (8 pts)
- Part 3: MLE and MAP (12 pts)
- Part 4: Naïve Bayes (10 pts)
- Part 5: Linear Regression (10 pts)
- Part 6: Logistic Regression and Neural Networks (10 pts)
- Part 7: SVM and Kernel (10 pts)

1 Multiple Choice Questions (20 pts)

There are 10 questions in this part, and each question worths 2 points. Write your answer in the box below each question.

Please select only one choice for each question.

1. Which of the following statements is **FALSE** about the Perceptron algorithm we introduced in the lecture?
- (A) The Perceptron algorithm is used to train a binary linear classifier.
 - (B) The Perceptron algorithm only works if the training data is linearly separable (*linearly separable* means that there exists a hyperplane that can separate the two classes without any point lying on the hyperplane itself).
 - (C) The Perceptron algorithm can always terminate if the data is linearly separable.
 - ☒ (D) The Perceptron algorithm can always find a linear classifier with the largest possible margin.

Your answer:

D

2. Given a linear classifier $y = \langle \mathbf{w}, \mathbf{x} \rangle$ (this notation means the dot product of vectors $\mathbf{w}$ and $\mathbf{x}$) where $\mathbf{w}$ is a **unit vector**, what is the margin of this classifier on the dataset $\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)\}$?
- (A) $\max_{i=1,2,\dots,n} |\langle \mathbf{w}, \mathbf{x}_i \rangle|$
 - ☒ (B) $\min_{i=1,2,\dots,n} |\langle \mathbf{w}, \mathbf{x}_i \rangle|$
 - (C) $\max_{i=1,2,\dots,n} |\langle \mathbf{w}, \mathbf{x}_i \rangle - y_i|$
 - (D) $\min_{i=1,2,\dots,n} |\langle \mathbf{w}, \mathbf{x}_i \rangle - y_i|$

Your answer:

B

3. What does the Occam's razor principle say about how to choose the best model?
- (A) Bigger models are always better than smaller models.
 - (B) Models with lower training error are always better than models with higher training error.
 - ☒ (C) The simplest model that is consistent with the data is the best model we should choose.
 - (D) To prevent overfitting, the number of parameters of the best model should be less than the number of training samples we have.

Your answer:

C

4. Which of the following statements is **TRUE** about the information gain?

- ☒ (A) The information gain can never be negative.
- ☐ (B) The information gain is equal to the expected increase in entropy.
- ☐ (C) The information gain is the highest for a binary target variable if it can divide the dataset into two subsets of equal size.
- ☐ (D) We can get the smallest possible decision tree if we choose the target variable with the highest information gain for each step.

Your answer:

A

5. Which of the following is an intuitive explanation of why the information gain criterion helps us choose a good split in a decision tree?

- ☐ (A) It measures the number of bits needed to encode the splitting rule.
- ☒ (B) It tends to make each leaf more certain about the correct class.
- ☐ (C) It tends to make the features conditionally independent within each leaf.
- ☐ (D) It tends to make the leaves have almost equal sizes.

Your answer:

B

6. In linear regression, with training data $\{(\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(n)}, y^{(n)})\}$, we want to find a $\mathbf{w}$ that minimizes the residual sum of squares (RSS) which can be written as

$$RSS(\mathbf{w}) = \sum_{i=1}^n \left(y^{(i)} - \left(w_0 + \sum_{j=1}^d w_j x_j^{(i)} \right) \right)^2$$

where the number of samples is n and the data dimension (dimension of $\mathbf{x}$) is d .

Which of the following statements is **TRUE** about RSS?

- ☒ (A) There is always a unique $\mathbf{w}$ that minimizes $RSS(\mathbf{w})$ if $n > d$.
- ☐ (B) In some cases it is possible that there is no finite $\mathbf{w}$ that minimizes $RSS(\mathbf{w})$.
- ☐ (C) For any $\mathbf{w}$ and $\mathbf{w}'$, we have $RSS(0.5\mathbf{w} + 0.5\mathbf{w}') \leq 0.5 \cdot RSS(\mathbf{w}) + 0.5 \cdot RSS(\mathbf{w}')$.
- ☐ (D) For RSS we always assume that each $y^{(i)}$ is either 1 or -1.

Your answer:

A

7. Which of the following best describes a GAN (generative adversarial network)?

- (A) It is used to attack a neural network by adding small noise to the input.
- (B) It is used to detect which distribution a certain input is sampled from.
- (C) It is trained by matching the learned representations of two neural networks.
- (D) It is trained by playing a minimax game between two neural networks.

Your answer:

D

8. The following are an English sentence and a Japanese sentence of the same meaning:

English: Welcome to 10-701!

Japanese: 10-701へようこそ!

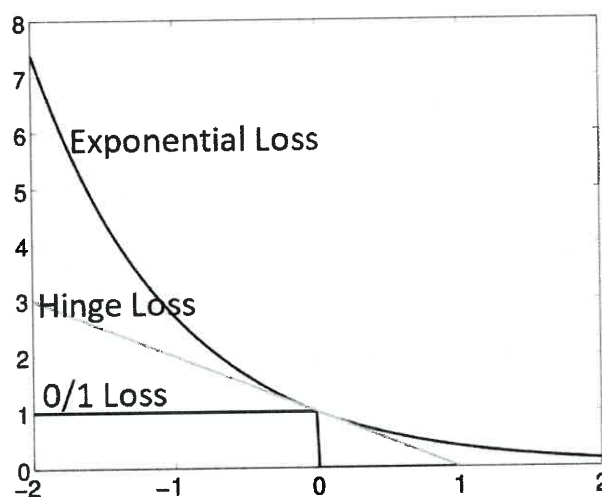
You can see that these two languages are very different. In Japanese, “へ” means “to” and “ようこそ” means “welcome”, so the Japanese sentence is actually “10-701 to welcome”. In addition, the numbers of characters are different for the two sentences (meaning that the two strings have different lengths). Given these difficulties, which of the following models should you use to do English-Japanese translation?

- (A) A Transformer.
- (B) A GAN.
- (C) A convolutional neural network.
- (D) An SVM.

Your answer:

A

9. In the SVM lecture we introduced several *surrogate losses* of the 0/1 loss, such as the hinge loss and the exponential loss, as shown in the diagram below. Which of the following statements is **FALSE** about surrogate losses?



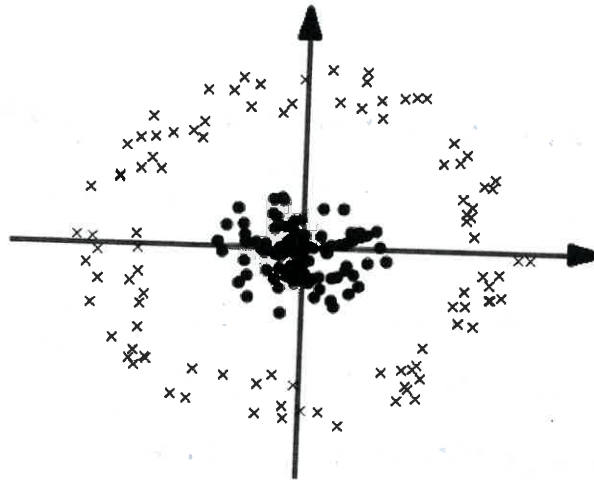
- (A) A surrogate loss should have some nice properties such as convexity and continuity so that it can be easily optimized.
- (B) A surrogate loss of the 0/1 loss should always be an upper bound of the 0/1 loss itself, that is it should never be less than the 0/1 loss.
- (C) The model with the lowest surrogate loss (of the 0/1 loss) on the training data is not necessarily the best model we should choose in practice.
- (D) We should use a surrogate loss to assess the performance of a model during test time because it is less sensitive to outliers than the 0/1 loss.

Your answer:

10. Recall from the lecture that a kernel is a function of two variables that can be written as the dot product of the same feature map of these variables:

$$K(x, x') = \phi(x)^\top \phi(x')$$

Now given a dataset as shown in the diagram below whose inputs are 2-dimensional, which kernel should you use so that you can train an SVM that separates these two classes (the dots • and the crosses ×) without making any mistake?



- (A) $K(x, x') = \|x - x'\|^2$
- (B) $K(x, x') = \|x\| \cdot \|x'\|$
- (C) $K(x, x') = \langle x, x' \rangle$ (this notation means the dot product)
- (D) $K(x, x') = \|x \odot x'\|$ ($\odot$ means element-wise multiplication of two vectors. For example, $(1, 2, 3) \odot (4, 5, 6) = (4, 10, 18)$)

Your answer:

2 Decision Trees (8 pts)

The TAs of 10-701 want to figure out which of the following factors influences a student's grade the most: (i) Whether the student attends lectures; (ii) Whether the student submits homework; (iii) Whether the student comes to office hours.

To do this, the TAs collect the data of 8 randomly sampled students from 10-701 in the previous year. 5 of the students get A- or above and 3 are below A-, as shown below:

Lecture	Homework	Office Hours	A- or above?
Y	Y	Y	Yes
Y	Y	N	Yes
Y	N	Y	Yes
Y	N	N	No
N	Y	Y	Yes
N	Y	N	Yes
N	N	Y	No
N	N	N	No

1
2
3
4
5
1

$\oplus$
 $\frac{5}{8}$
 $\ominus$
 $\frac{3}{8}$

Now, the TAs want you to help them build a decision tree to predict if a student can get at least A-.

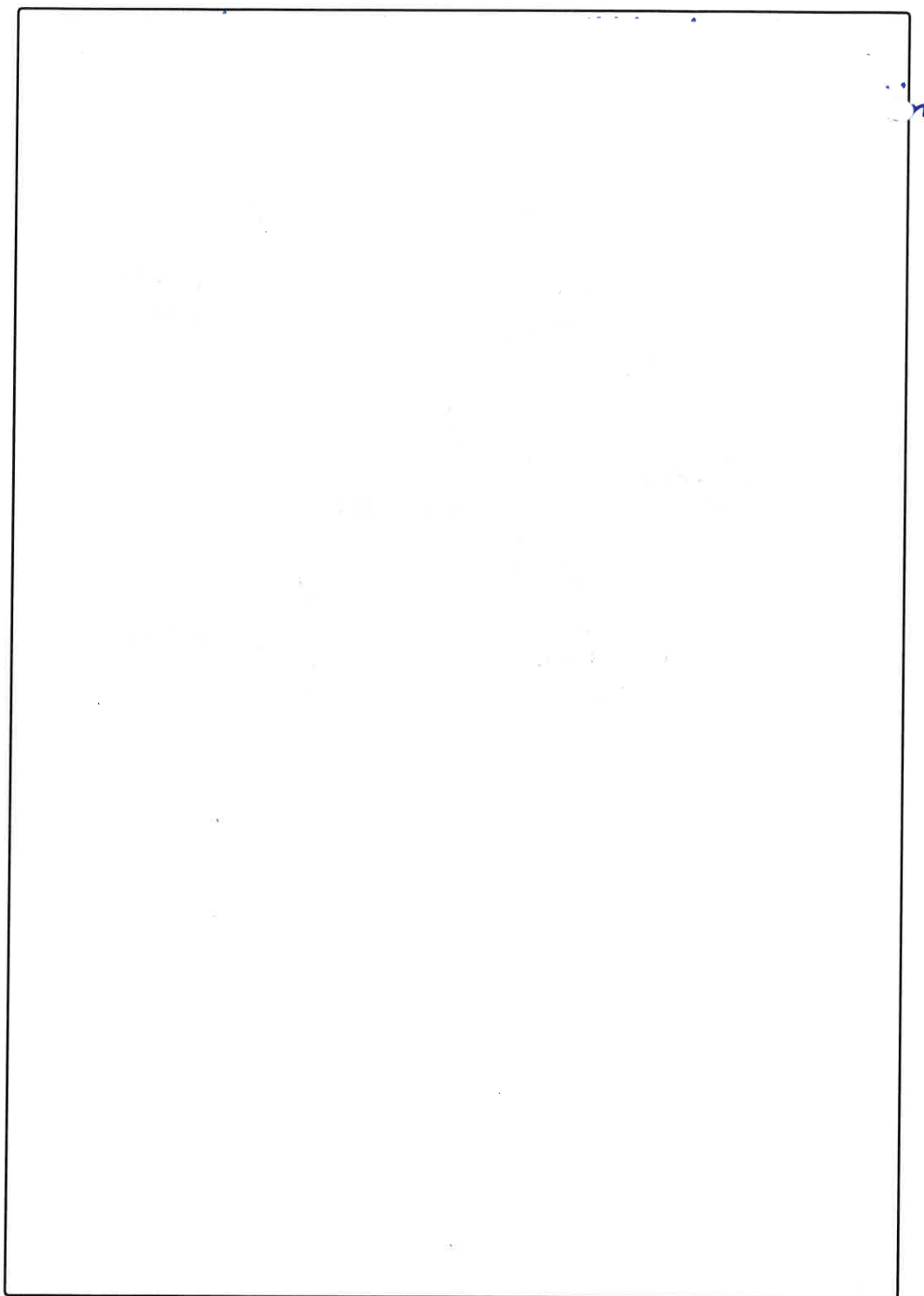
- (6 pts) Which of the three factors has the highest information gain so that you should put it in the root of the tree with the ID3 algorithm? Justify your answer.

Hint: The entropy of random variable X is $H(X) = -\sum_x \mathbb{P}(X=x) \log_2 \mathbb{P}(X=x)$.

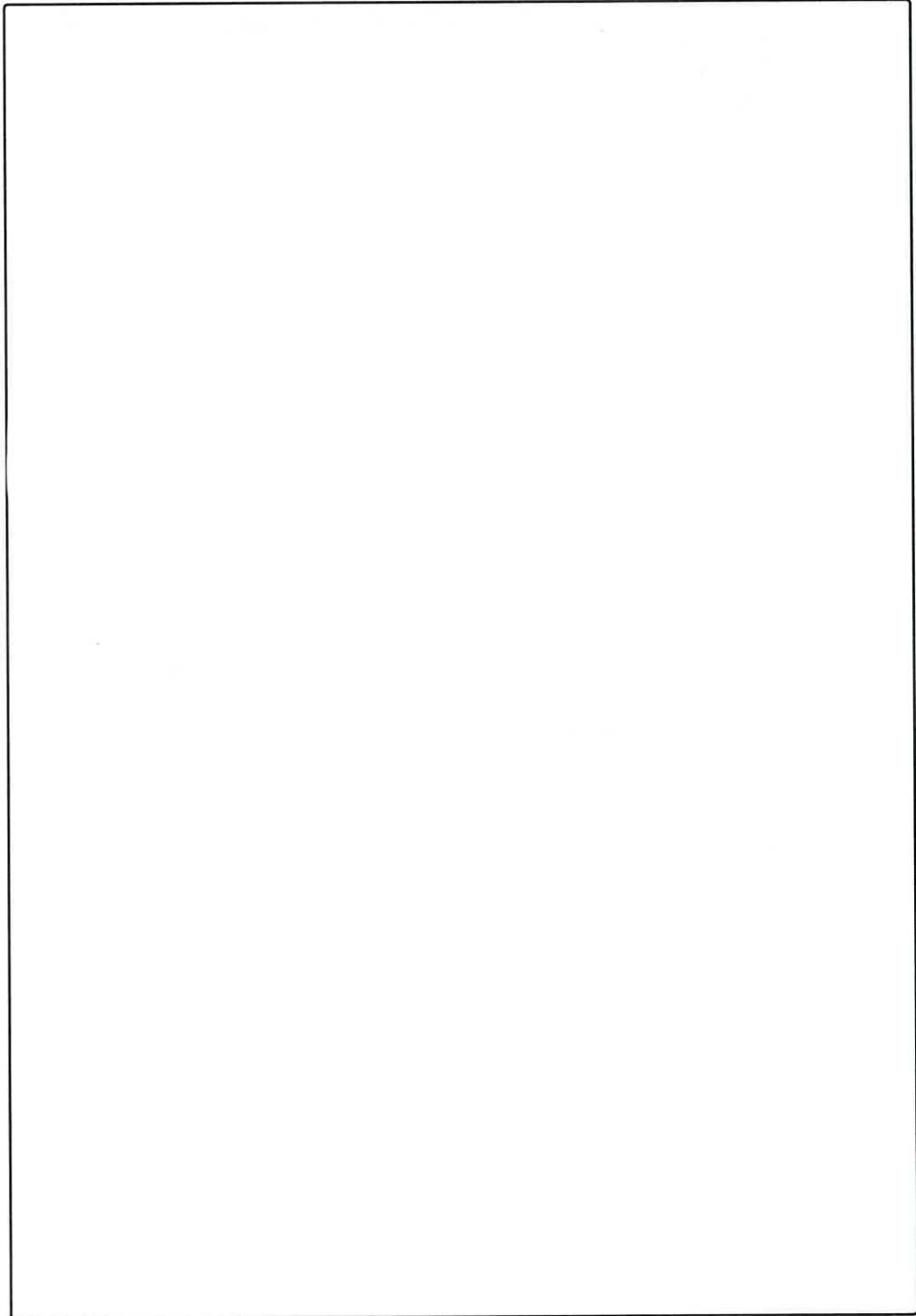
$$H(S) = -P_{\oplus} \log_2 P_{\oplus} - P_{\ominus} \log_2 P_{\ominus}$$

$$H(S) = -\frac{5}{8} \log_2 \left(\frac{5}{8}\right) - \frac{3}{8} \log_2 \left(\frac{3}{8}\right) = E$$

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2. (2 pts) Now, based on your answer to the previous question, complete the decision tree. You just need to draw the tree. No explanation is needed.



3 MLE and MAP (12 pts)

1. (4 pts) Let $X_1, X_2, \dots, X_N$ be non-negative integer-valued random variables independently drawn from a Poisson distribution with parameter λ . Recall that a Poisson distribution is a discrete probability distribution over non-negative integers k , such that

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}.$$

You can use without proof the following property of the Poisson distribution in this part: Its mean is λ , meaning that $\mathbb{E}[X] = \lambda$ if X is sampled from the Poisson distribution.

Given observations $X_1 = k_1, X_2 = k_2, \dots, X_N = k_N$, show that the Maximum Likelihood Estimate (MLE) is equal to the mean of the observations: $\hat{\lambda}^{MLE} = \sum_{i=1}^N k_i / N$.

2. (2 pts) An estimator $\hat{\lambda}$ is called *unbiased* if it satisfies the following equation:

$$\mathbb{E}[\hat{\lambda}(X_1, \dots, X_n)] = \lambda$$

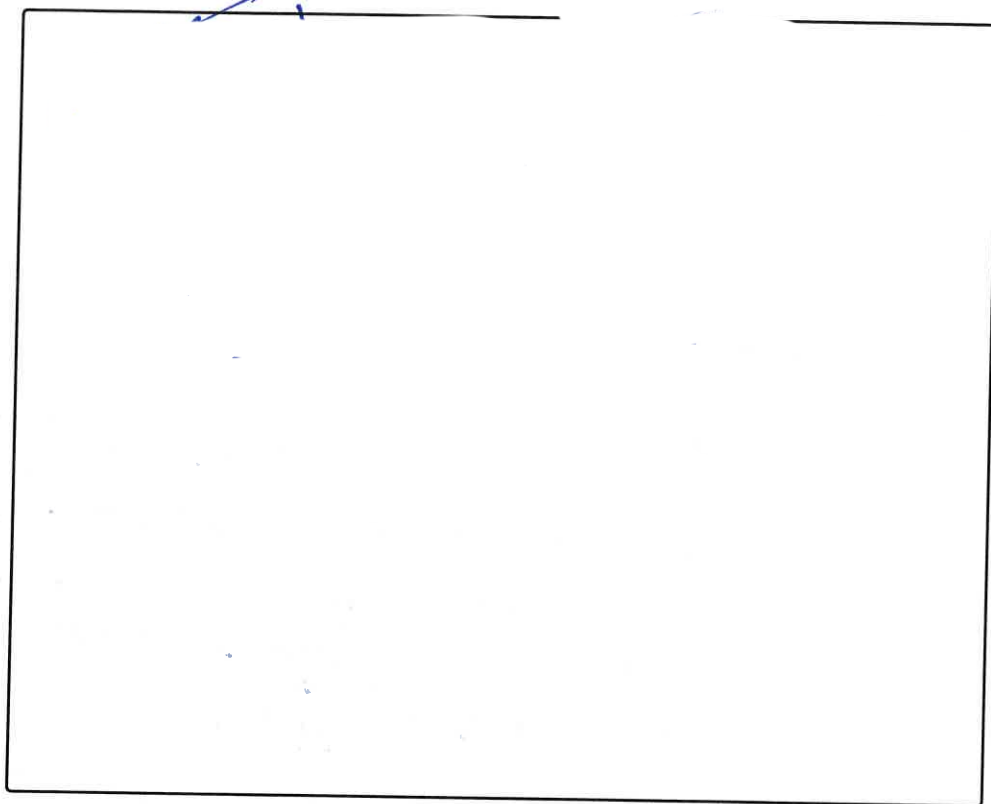
Here we view $\hat{\lambda}$ as a function of $X_1, \dots, X_n$, λ is the ground truth parameter of the distribution from which $X_1, \dots, X_n$ are sampled, and the expectation is taken over the random sampling of $X_1, \dots, X_n$. Question: Is the above estimate $\hat{\lambda}^{MLE}$ unbiased? Briefly justify.

3. (4 pts) Under the same conditions as in question 1, now additionally suppose that the prior distribution of λ is the Gamma distribution with parameters $\alpha > 0$ and $\beta > 0$. The probability density function of λ is given by

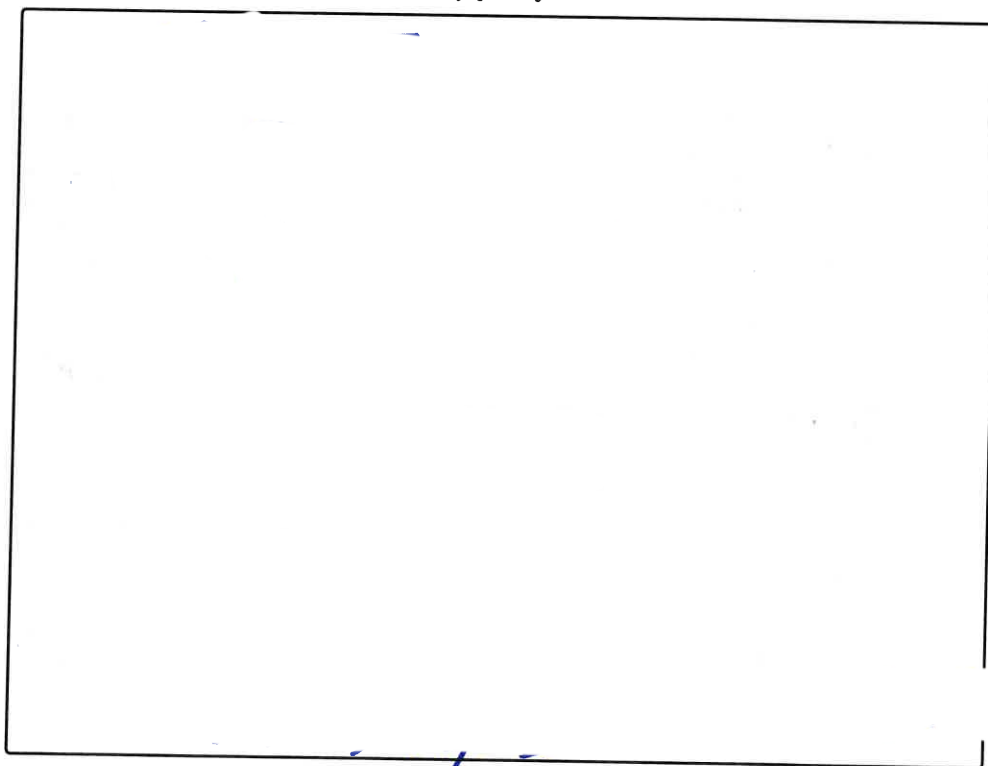
$$p(\lambda) = C\lambda^{\alpha-1}e^{-\beta\lambda},$$

for $\lambda \in (0, \infty)$ where C is a normalization constant. Given observations $X_1 = k_1, X_2 = k_2, \dots, X_N = k_N$, what is the MAP estimate $\hat{\lambda}^{MAP}$? Show your work.

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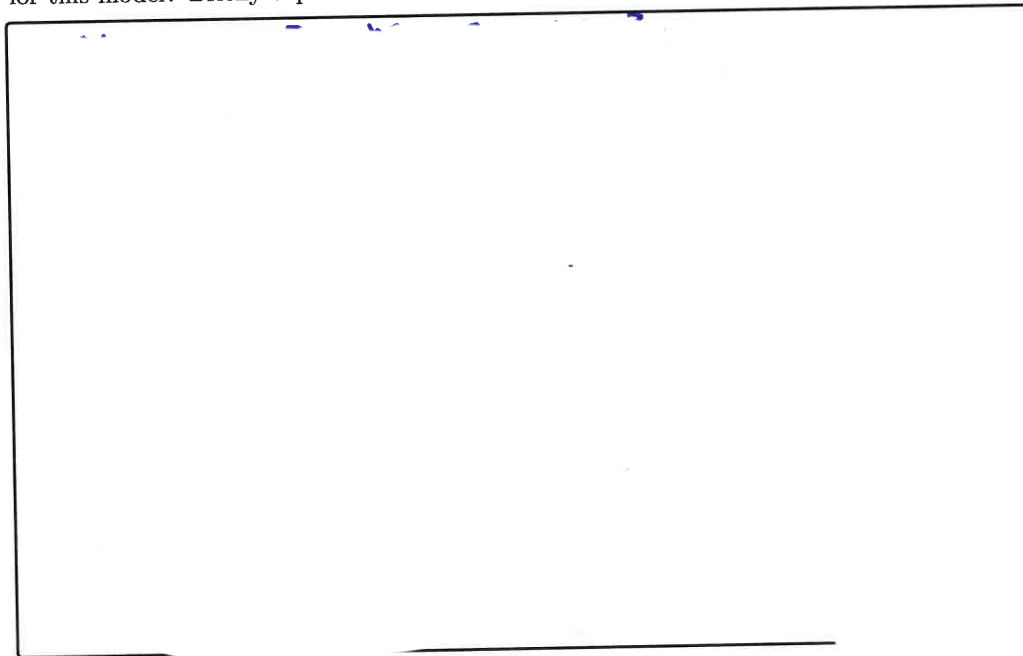


4. (2 pts) Is the MAP estimator you got in the previous question $\hat{\lambda}^{MAP}$ unbiased for arbitrary choices of α and β ? Briefly justify.

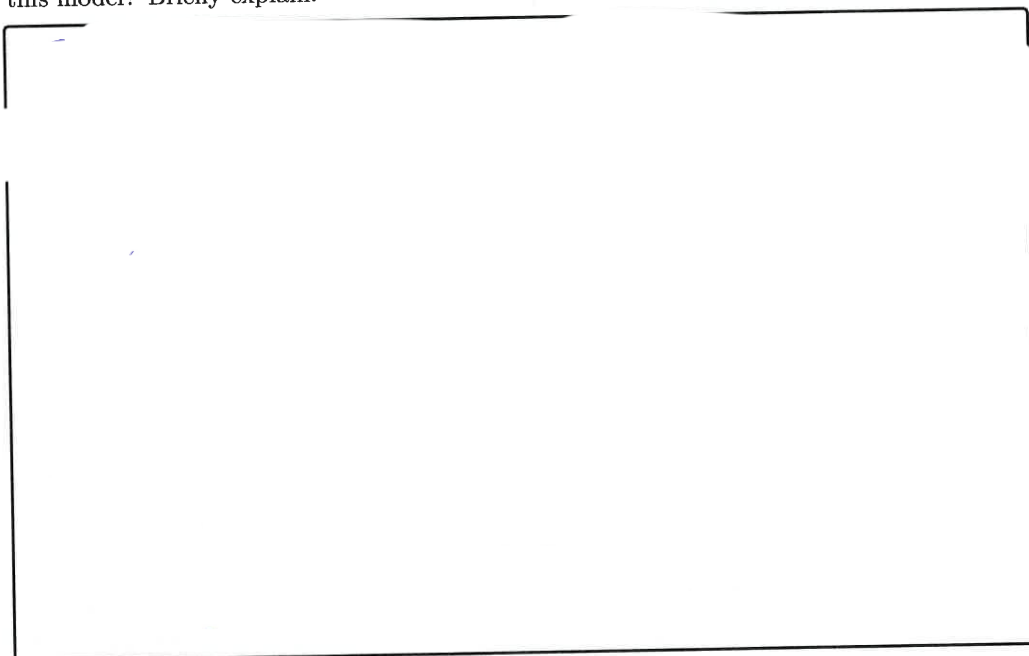


4 Naïve Bayes (10 pts)

1. (2 pts) Consider training a Bayes model $P(Y = y|X) \propto P(X|Y = y)P(Y = y)$ to estimate $P(Y|X_1 \dots X_n)$, where $Y \in \{0, 1, 2\}$, and $X_i \in \{0, 1, 2, 3, 4\}$ for all i . With the Naïve Bayes assumption, how many independent parameters do you need at least for this model? Briefly explain.



2. (2 pts) Under the same settings as the previous question, but this time without the Naïve Bayes assumption. How many independent parameters do you need at least for this model? Briefly explain.



Now, for the remaining questions in this part, suppose you train a Bayes classifier with the Naïve Bayes assumption on the dataset in Table 1, where each example consists of 3 features. In your model, each feature x_i is sampled from a categorical distribution where $x_i \in \{a, b, c\}$. The label y is sampled from a Bernoulli distribution where $y \in \{0, 1\}$.

Table 1: Dataset

x_1	x_2	x_3	y
a	b	c	1
b	c	c	1
c	c	a	1
a	c	c	0
a	a	b	1

Table 2: Test Example

x_1	x_2	x_3	y
a	c	c	

Table 3: Maximum Likelihood Estimates

$p(y = 0)$	
$p(y = 1)$	
$p(x_1 = a \mid y = 0)$	
$p(x_1 = a \mid y = 1)$	
$p(x_2 = c \mid y = 0)$	
$p(x_2 = c \mid y = 1)$	
$p(x_3 = c \mid y = 0)$	
$p(x_3 = c \mid y = 1)$	

- (4 pts) Compute the maximum likelihood estimates of the Naive Bayes parameters requested in Table 3. **Write your answers in the table.**
- (2 pts) How would your Bayes classifier in the previous question label the example in Table 2 at test time? Justify your answer.

5 Linear Regression (10 pts)

Consider the linear regression setup, with data $\mathcal{D} = \{(x_1, y_1), \dots, (x_n, y_n)\}$, $x_i \in \mathbb{R}$, $y_i \in \mathbb{R}$ generated by a true model of form

$$Y_i = \beta_0^* + \beta_1^* X_i + \epsilon$$

Here $\beta_0^* \in \mathbb{R}$, $\beta_1^* \in \mathbb{R}$ are two parameters, and $\epsilon \sim N(0, \sigma^2)$ is random noise. In least squares regression we aim to find estimators $(\hat{\beta}_0, \hat{\beta}_1)$ that minimize residual sum of squares (RSS),

$$\underset{\beta_0, \beta_1}{\operatorname{argmin}} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2. \quad (1)$$

We further assume that β_0^* is zero, so we can simplify Equation (1) to

$$\hat{\beta}_1 = \underset{\beta_1}{\operatorname{argmin}} \sum_{i=1}^n (y_i - \beta_1 x_i)^2. \quad (2)$$

1. (4 pts) Derive the expression for $\hat{\beta}_1$ in Equation (2). Show your work.

2. (3 pts) In the above setting, it can be shown that the random variable Y_i , given $X_i = x_i$, follows the normal distribution $N(\beta_1^* x_i, \sigma^2)$, with probability density function

$$\mathbb{P}(Y_i = z | X_i = x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(z - \beta_1^* x_i)^2}$$

An alternative way to estimate β_1^* is to use the maximum likelihood estimator $\hat{\beta}_{\text{MLE}}$, which can be computed by maximizing the log-likelihood of the data $\mathcal{D}$. What is the log-likelihood of observations $\mathcal{D} = \{(x_1, y_1), \dots, (x_n, y_n)\}$?

Hint: The log-likelihood in this case is: $\log \mathbb{P}(Y_1, \dots, Y_n | X_1 = x_1, \dots, X_n = x_n)$.

3. (3 pts) Argue that $\hat{\beta}_{MLE}$ obtained by maximizing the log-likelihood in the previous question, is the same as $\hat{\beta}_1$ obtained by minimizing the squared error in Equation (2).

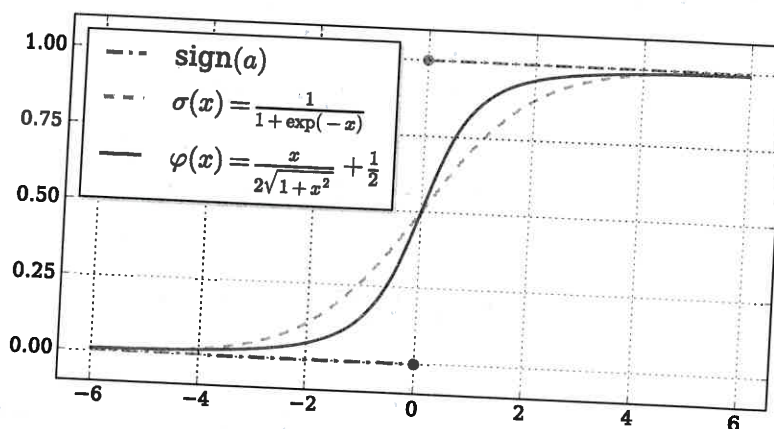
6 Logistic Regression and Neural Networks (10 pts)

In binary classification, a linear classifier is defined as the sign of a linear function: $\text{sign}(\mathbf{w}^\top \mathbf{x} + b)$. However, the sign function is not differentiable, so it is not suitable for training. To solve this problem, in logistic regression we use the sigmoid function $\sigma(a)$ to compute the probability of a sample $\mathbf{x}$ having label $y = 1$.

One issue of the sigmoid function, however, is that it is somewhat shallow around zero so that it cannot approximate the sign function well enough. Thus, in this problem we will investigate an alternative called the *steeper sigmoid*, which is defined as

$$\varphi(a) = \frac{a}{2\sqrt{1+a^2}} + \frac{1}{2}$$

The following diagram demonstrates the difference between sigmoid and steeper sigmoid:



(Note: In this plot the sign function takes values in $\{0, 1\}$, different from the conventional definition. You don't need to use the sign function in this problem.)

You can observe from the plot that the steeper sigmoid is indeed steeper around zero.

Armed with this new function, we can define a probability distribution as follows:

$$\mathbb{P}_{\mathbf{w},b}(y = 1|\mathbf{x}) = \varphi(\mathbf{w}^\top \mathbf{x} + b) \quad \text{and} \quad \mathbb{P}_{\mathbf{w},b}(y = -1|\mathbf{x}) = 1 - \mathbb{P}_{\mathbf{w},b}(y = 1|\mathbf{x})$$

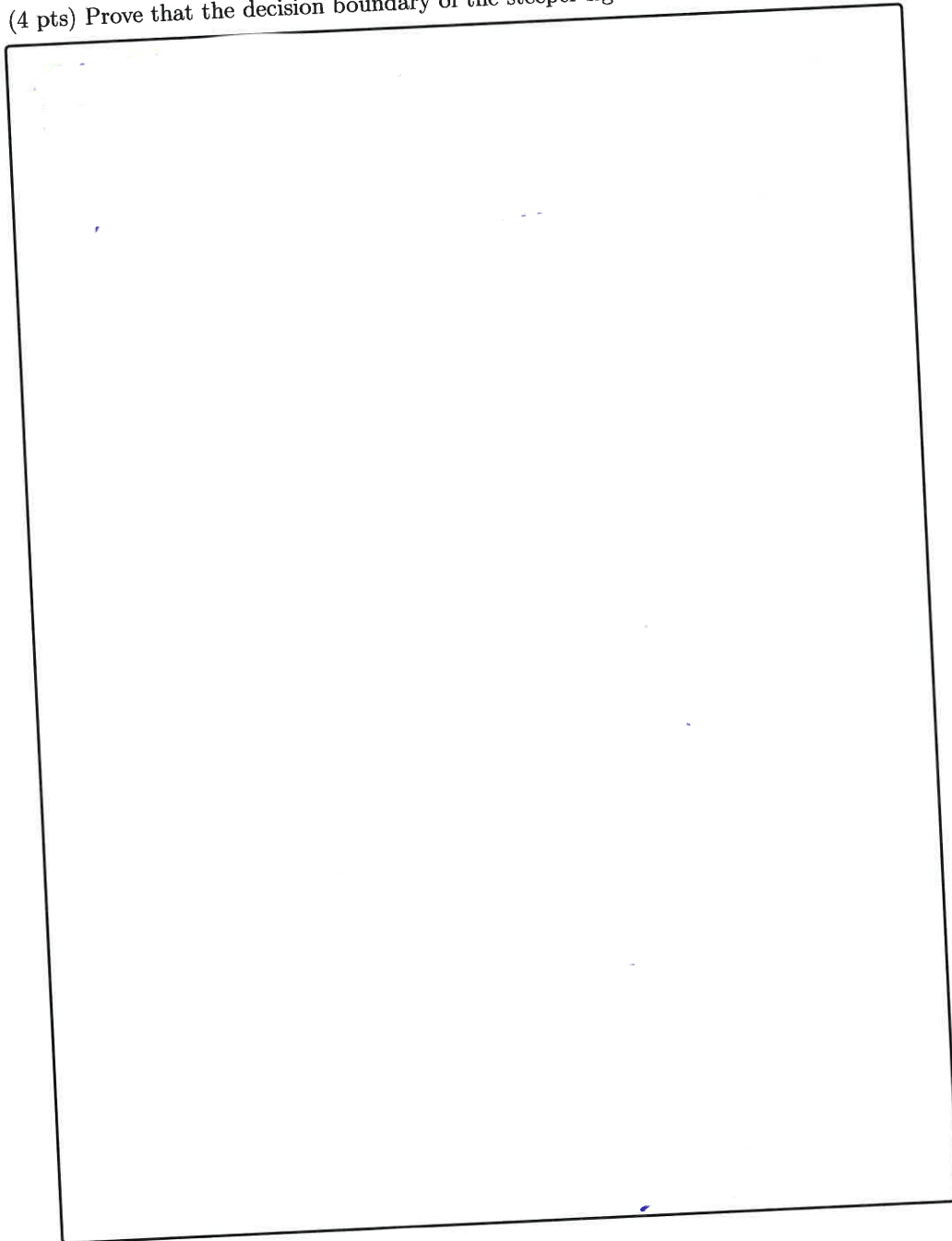
And the Bayes classifier associated with this probability distribution is:

$$h_{\mathbf{w},b}(\mathbf{x}) = \underset{k \in \{-1, 1\}}{\text{argmax}} \mathbb{P}_{\mathbf{w},b}(y = k|\mathbf{x})$$

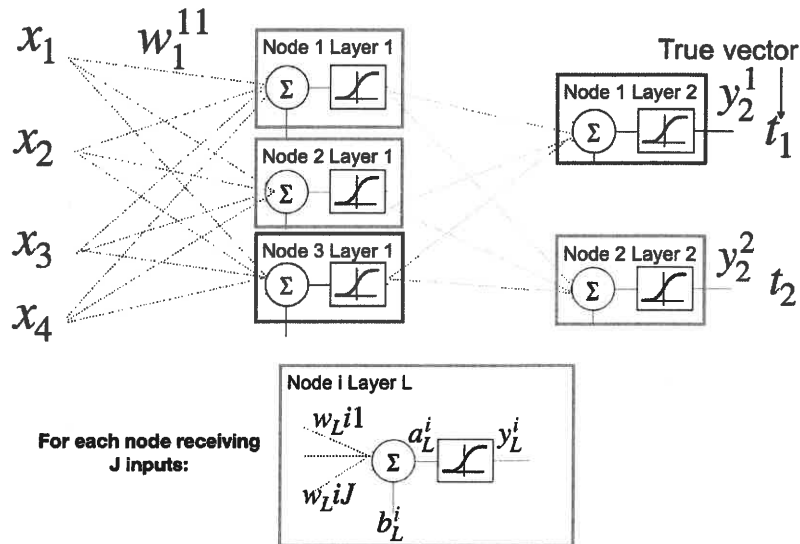
We will call this Bayes classifier the *steeper sigmoid classifier*.

Also recall that the *decision boundary* of a binary classifier is the set of all $\mathbf{x}$ where the probability of the prediction y being 1 and -1 are the same.

1. (4 pts) Prove that the decision boundary of the steeper sigmoid classifier is linear.



2. (4 pts) We now use this new function to build the following neural network:



Here w_L^{ij} is the parameter (a real value) of the edge that goes from neuron j in the $(L - 1)$ -th layer to neuron i in the L -th layer. a_L^i is the output before activation in layer L neuron i , and y_L^i is the output after activation in layer L neuron i . The zeroth layer is the input layer, which accepts a 4-dimensional input (x_1, x_2, x_3, x_4) . The first layer is a 3-dimensional hidden layer, and the second layer is a 2-dimensional output layer that outputs (y_2^1, y_2^2) . Note that the activation is also applied in the last layer. The activation function is the steeper sigmoid φ we defined in the previous question.

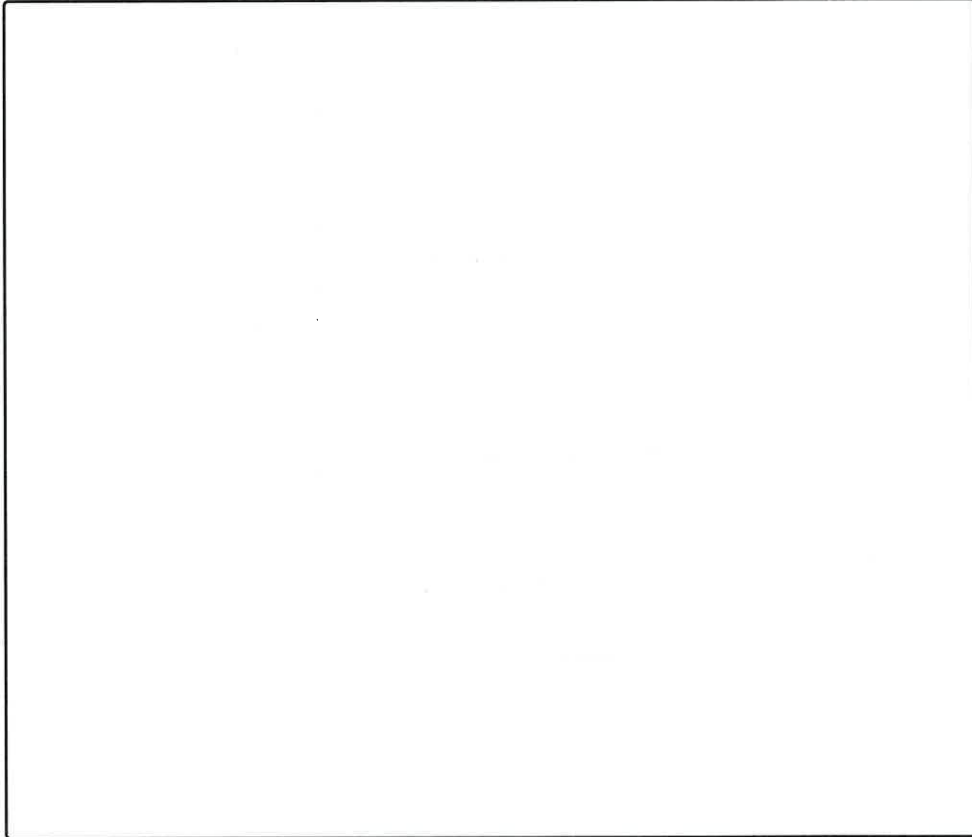
We train this neural network by minimizing the following mean squared error:

$$E(\hat{y}, t) = \frac{1}{2} [(y_2^1 - t_1)^2 + (y_2^2 - t_2)^2]$$

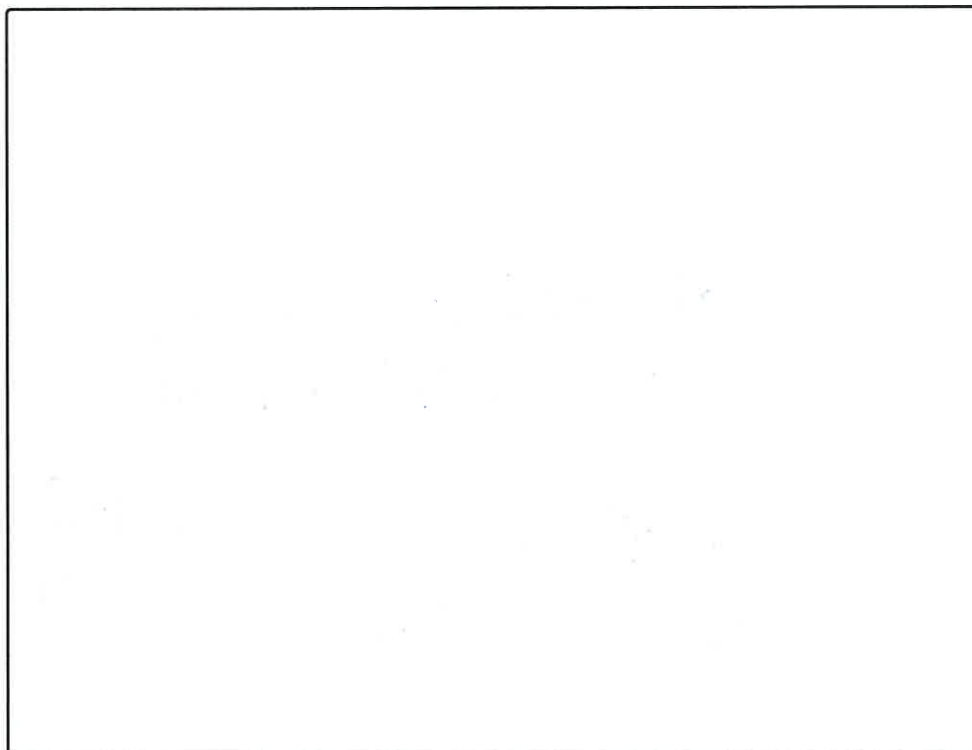
where (y_2^1, y_2^2) is the output of the network, and (t_1, t_2) is the ground truth.

Question: What is $\frac{\partial E}{\partial w_1^{11}}$? Please keep the derivative of φ as $\varphi'(x)$ and do not expand it. Your formula can include a_L^i , y_L^i , x_i , t_i and φ' .

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3. (2 pts) How will the decision boundary of this neural network be different than that of the steeper sigmoid classifier?



7 SVM and Kernel (10 pts)

Recall from the lecture that the primal optimization problem for SVM is:

$$\begin{aligned} \min_{\mathbf{w}, b} \quad & \frac{1}{2} \|\mathbf{w}\|^2 \\ \text{s.t.} \quad & y_i(\langle \mathbf{x}_i, \mathbf{w} \rangle + b) \geq 1, \quad i = 1, \dots, m \end{aligned}$$

where $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m)\}$ is the training dataset. Also recall from the lecture that the dual of this problem is:

$$\begin{aligned} \max_{\alpha_1, \dots, \alpha_m} \quad & \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m \alpha_i \alpha_j y_i y_j \langle \mathbf{x}_i, \mathbf{x}_j \rangle \\ \text{s.t.} \quad & \alpha_1, \dots, \alpha_m \geq 0 \\ & \sum_{i=1}^m \alpha_i y_i = 0 \end{aligned}$$

where the optimal weight $\mathbf{w}$ and the formula of the classifier $F(\mathbf{x})$ are given by

$$\mathbf{w} = \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i \quad F(\mathbf{x}) = \text{sign}(\langle \mathbf{w}, \mathbf{x} \rangle + b) = \text{sign}\left(\sum_{i=1}^m \alpha_i y_i \langle \mathbf{x}_i, \mathbf{x} \rangle + b\right)$$

Moreover, with the kernel trick, the dual problem of *kernel SVM* becomes

$$\begin{aligned} \max_{\alpha_1, \dots, \alpha_m} \quad & \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j) \\ \text{s.t.} \quad & \alpha_1, \dots, \alpha_m \geq 0 \\ & \sum_{i=1}^m \alpha_i y_i = 0 \end{aligned}$$

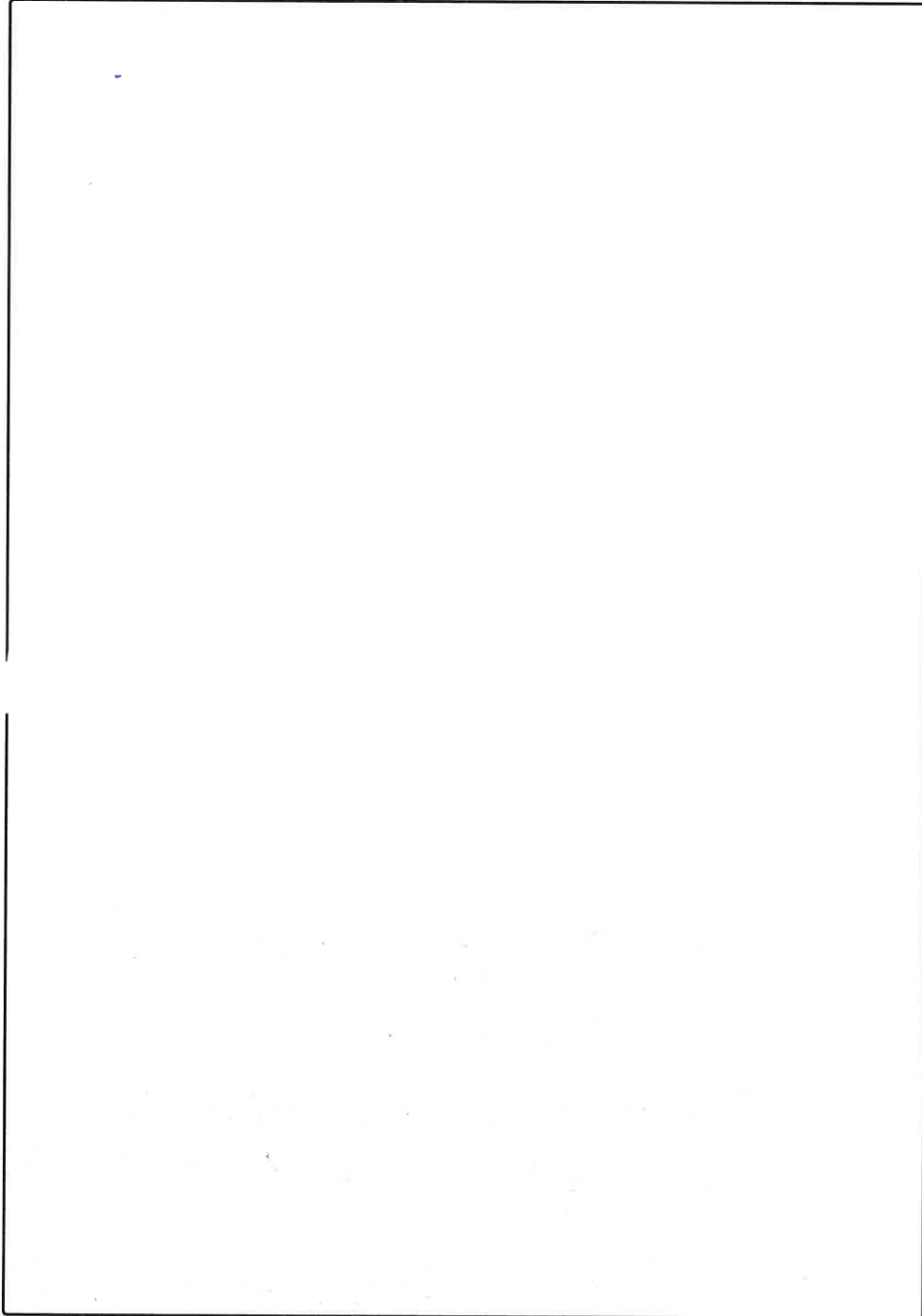
Also recall that a kernel is any function that can be written as $K(\mathbf{x}_i, \mathbf{x}_j) = \langle \Phi(\mathbf{x}_i), \Phi(\mathbf{x}_j) \rangle$ where Φ is a feature map.

1. (2 pts) What is the formula of $F(\mathbf{x})$ for the kernel SVM? Briefly explain.

2. (3 pts) Show that if $K(\mathbf{x}_i, \mathbf{x}_j) : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a kernel, then for any positive function $\alpha : \mathcal{X} \rightarrow \mathbb{R}^+$,

$$K'(\mathbf{x}_i, \mathbf{x}_j) = \frac{K(\mathbf{x}_i, \mathbf{x}_j)}{\alpha(\mathbf{x}_i)\alpha(\mathbf{x}_j)}$$

is also a kernel.

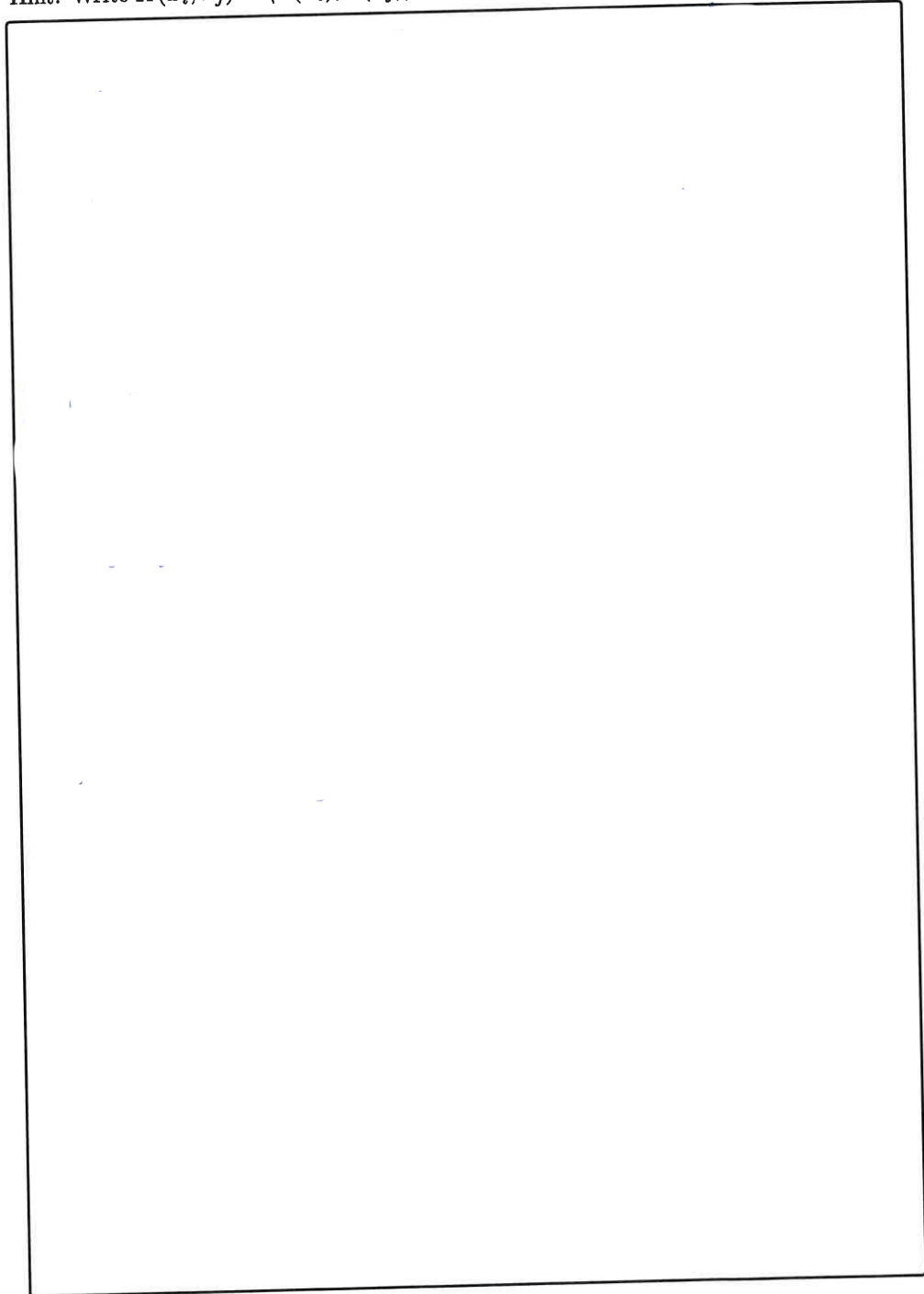


3. (5 pts) Show that if $K(\mathbf{x}_i, \mathbf{x}_j) : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a kernel, then for any **non-negative** real number c and any positive integer d ,

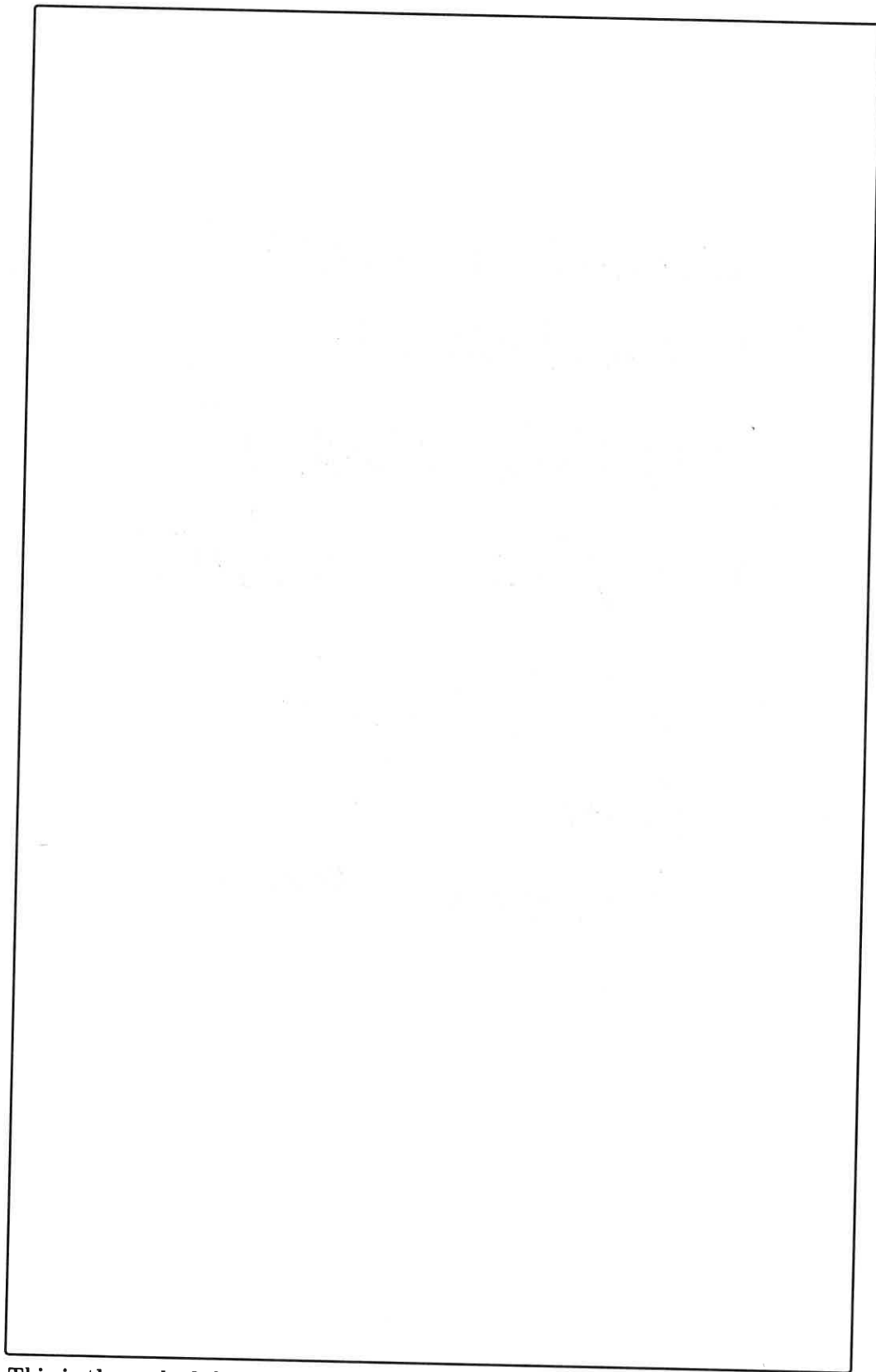
$$K'(\mathbf{x}_i, \mathbf{x}_j) = [K(\mathbf{x}_i, \mathbf{x}_j) + c]^d$$

is also a kernel (which is called the *polynomial kernel*).

Hint: Write $K(\mathbf{x}_i, \mathbf{x}_j) = \langle \Phi(\mathbf{x}_i), \Phi(\mathbf{x}_j) \rangle$ and expand, and then try to find a Φ' for K' .



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This is the end of the exam.